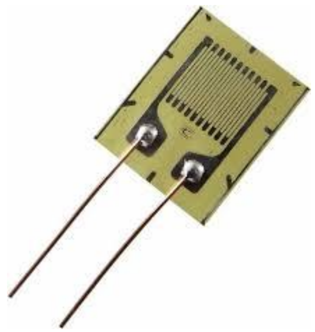
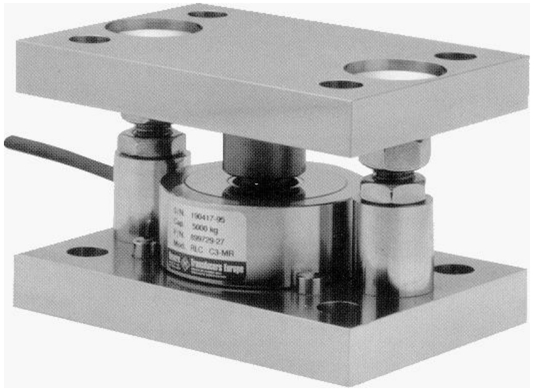
درس ابزار دقیق

حسگرهای پیزورزیستیو و لودسل





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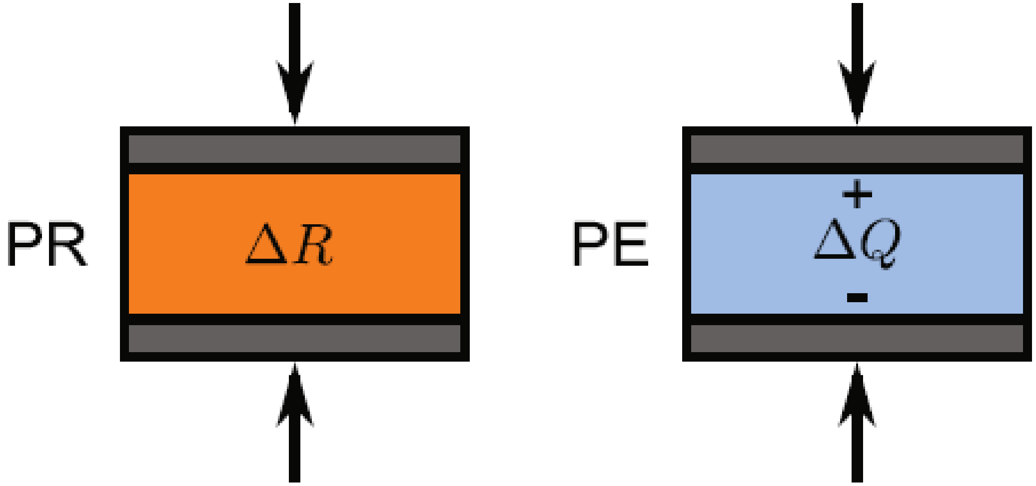
اردیبهشت ماه ۱۴۰۳

فهرست مطالب

- معرفی پدیده پزورزیستیو و مقایسه با پدیده پیزوالکتریک
- علت تغییر در مقاومت با اعمال نیرو در فلزات و نیم رساناها
- معرفی ماتریس پیزورزیستو
- انواع پیزورزیستورها
- پارامترهای مهم پیزورزیستورها
- کاربردهای حسگر پیزورزیستو
- مثالهایی از Loadcell ها و کاربردها

- در مواد پیزوالکتریک با اعمال نیرو تغییر در پلاریزاسیون الکتریکی اتفاق می افتد و در ساختارهای شبیه به خازن این تغییر پلاریزاسیون باعث ایجاد بار در صفحات هادی دو در دو طرف و همچنین ایجاد ولتاژ در صفحات می شود. در حالی که در مواد پیزورزیستیو اعمال نیرو به ماده باعث تغییر در مقاومت الکتریکی آن می شود.



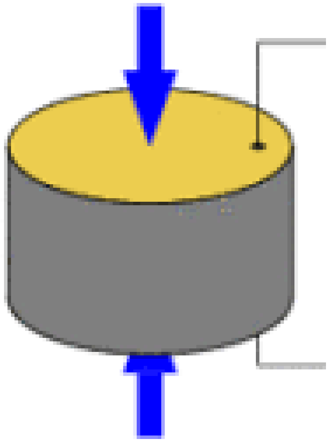
Mechanical stress induces a resistance change in piezoresistive (PR) materials and charge polarization in piezoelectric (PE) materials.

The fundamental principle of piezoresistive sensors comes from Equation:

$$R = \rho \frac{l}{A}$$

Taking natural logarithm on both sides yields:

$$\ln(R) = \ln(\rho) + \ln(l) - \ln(A)$$



The differential of the aforementioned equation becomes:

$$\frac{dR}{R} = \frac{d\rho}{\rho} + \frac{dl}{l} - \frac{dA}{A}$$

Let $A = \pi D^2/4$, where D is the diameter of a wire:

$$\ln(A) = \ln(\pi/4) + 2\ln(D)$$

The differential of this equation results:

$$\frac{dA}{A} = 2\frac{dD}{D}$$

where $dD/D = \varepsilon_D$ is the *transverse* or *lateral strain*. Since the *longitudinal strain* is $dl/l = \varepsilon$, and *Poisson's ratio* is $\upsilon = -\varepsilon_D/\varepsilon$, Equation 2.42 becomes:

$$\frac{dR}{R} = \frac{d\rho}{\rho} + (1 + 2\upsilon)\varepsilon \qquad (2.43)$$

Equation 2.43 expresses the basic relationship between resistance and strain. A measure of the sensitivity of the material (i.e., its resistance change per unit of applied strain) is defined as the *gauge factor*:

$$\text{Gauge factor (GF)} = \frac{dR/R}{\varepsilon} \qquad (2.44)$$

Thus, the GF of Equation 2.43 is

$$\text{GF} = (1 + 2\upsilon) + \frac{d\rho/\rho}{\varepsilon} \qquad (2.45)$$

EXAMPLE 2.16

A gauge has GF 2.0 and resistance 120 Ω. Find dR when the gauge is subjected to a strain of (1) 5 microstrain in aluminum and (2) 5000 microstrain in aluminum.

SOLUTIONS

According to Equation 2.44

1. $dR = \mathrm{GF}\ \varepsilon\ R = 2\ (5 \times 10^{-6})\ (120\ \Omega) \Rightarrow 0.0012\ \Omega$ (0.001% change)
2. $dR = \mathrm{GF}\ \varepsilon\ R = 2\ (5000 \times 10^{-6})\ (120\ \Omega) = 1.2\ \Omega$ (1% change)

Piezoresistive effect in silicon and germanium was discovered by Charles Smith in 1954. He found that both *p*-type and *n*-type silicon and germanium exhibited much greater piezoresistive effect than metals.

Piezoresistivity of silicon arises from the **deformation of the energy bands** as a result of applied stress. In turn, the deformed bands affect the effective mass and the mobility of electrons and holes, hence modifying **resistivity** or conductivity.

- در نیم رساناها اثر تغییر در مقاومت ویژه ρ با اعمال نیرو نسبت به اثراتی که نیرو در تغییر ابعاد ایجاد می کنند بسیار بیشتر است.

In practical applications, a thin strip of silicon is commonly used to make a strain gauge sensor, instead of a three-dimensional cube. In this case, change in resistance versus in-plane stresses in the longitudinal (parallel to the current) direction and transverse (perpendicular to the current) direction can be expressed as

$$\frac{\Delta R}{R} = \pi_L \sigma_L + \pi_T \sigma_T \tag{2.48}$$

where ΔR and R are the change in resistance and the original resistance, respectively; σ_L and σ_T are the longitudinal and transverse stress, respectively; π_L and π_T are the piezoresistive coefficient along the longitudinal and transverse direction, respectively.

EXAMPLE 2.17

A pressure sensor die contains four identical *p*-type piezoresistors (Figure 2.25). Resistors *A* and *C* are subjected to the longitudinal stress σ_L, and Resistors *B* and *D* are subjected to the transverse stress component σ_T. Assume that the square-shaped diaphragm is under uniform pressure loading at the top surface, $\sigma_L = \sigma_T = 186.8$ MPa, $\pi_L = \pi_T = 2.762 \times 10^{-11}$ Pa^{-1}. Estimate $\Delta R/R$.

Solution

Since the square diaphragm is subjected to a uniform pressure load at the top surface, the bending moments are normal to all piezoresistors and are equal in magnitudes. Thus, for each resistor

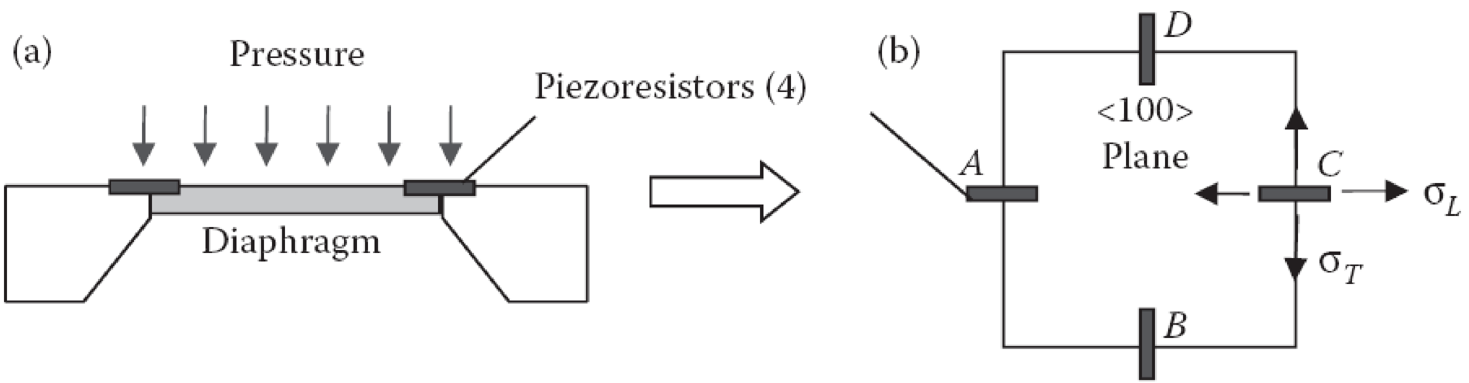


FIGURE 2.25 A semiconductor pressure sensor die: (a) side view; (b) top view.

$$\frac{\Delta R}{R} = \pi_L\sigma_L + \pi_T\sigma_T = 2 \times (2.762 \times 10^{-11}\ \text{Pa}^{-1})(186.8 \times 10^{6}\ \text{Pa}) = 0.01032 \text{ or } 1.032\%$$

2.5.4.1 Types and Structures of Strain Gauges

Types of strain gauges include wire strain gauges, foil strain gauges, single-crystal semiconductor strain gauges, and thin-film strain gauges.

2.5.4.1.1 Wire Strain Gauges

A wire strain gauge is the original type of resistive strain gauge. The first bonded, metallic wire-type strain gauges were developed in 1938. They are still extensively used in high-temperature environments today. The wire is made of metals or their alloys with a typical diameter of 6 ~ 30 μm. Common wire materials are CrNi alloys for standard applications and PtW alloys for high-temperature applications.

2.5.4.1.2 *Metal Foil Strain Gauges*

A foil strain gauge (see Figure 2.29) is the most widely used type. A very thin metal foil pattern (2~5 μm thick, usually Constantan or Nichrome V) is deposited onto a thin insulating backing or carrier (10~30 μm thick, usually epoxy, polyimide, or polycarbonate). The measuring grid pattern including the metallic terminal tabs is produced by the photoetching process. The entire gauge is typically 5~15 mm long.

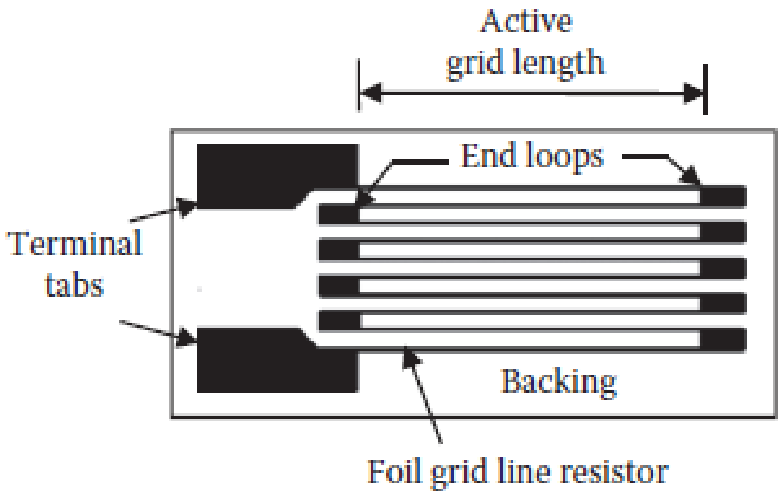


FIGURE 2.29 Metal foil strain gauge construction.

2.5.4.1.3 *Single-Crystal Semiconductor Strain Gauges*

This type of strain gauge is manufactured from a thin strip of semiconductor cut from a single crystal of silicon or germanium, doped with accurate amounts of impurities to obtain either *n*- or *p*-type. The output of a semiconductor gauge is very high compared to a wire or foil gauge. Semiconductor gauges can provide both positive and negative gauge factors depending on whether the gauges are *n*-type or *p*-type. The typical gauge factor of a semiconductor is −100 ~ +170, although −115 ~ +205 are achievable.

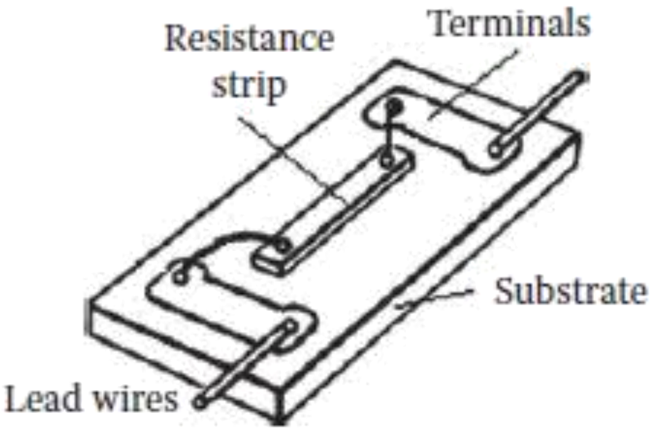


FIGURE 2.30 A single-crystal semiconductor strain gauge. (United States Patent 3,609,625.)

2.5.4.1.4 *Thin-Film Strain Gauges*

In thin-film strain gauges, the thin film (resistance element) is produced by sputtering or evaporating metals, alloys, or semiconductors (silicon or germanium) onto the carrier material (as a substrate) in vacuum. Several stages of sputtering and evaporation may be needed to form up to eight layers of material in thin-film polycrystalline structure. The whole surface is finally coated with an appropriate sealant. The final properties of the thin film are strongly influenced by various parameters of the deposition process (e.g., ultimate vacuum, substrate temperature, and rate of film growth). These processes are highly cost effective when strain gauges are produced in large quantities. The structure of a thin-film gauge is shown in Figure 2.31. The gauge factor of a thin-film strain gauge is higher than those of wire or foil gauges, but lower than those of single-crystal semiconductor gauges. For instance, a germanium thin-film strain gauge has a GF of 32–39.

The nominal resistance of each type of strain gauges is shown in Table 2.16.

TABLE 2.16
Nominal Resistance of Various Strain Gauges

Type of Strain Gauge	Nominal Resistance
Wire gauge	60~350 Ω
Foil or semiconductor gauge	120 Ω~5 kΩ
Thin-film gauge	~10 kΩ

پارامترهای مهم پیزورزیستورها

The characteristics of a strain gauge are mainly defined by:

Gauge dimensions and shape,

Resistance,

Gauge factor (Strain sensitivity),

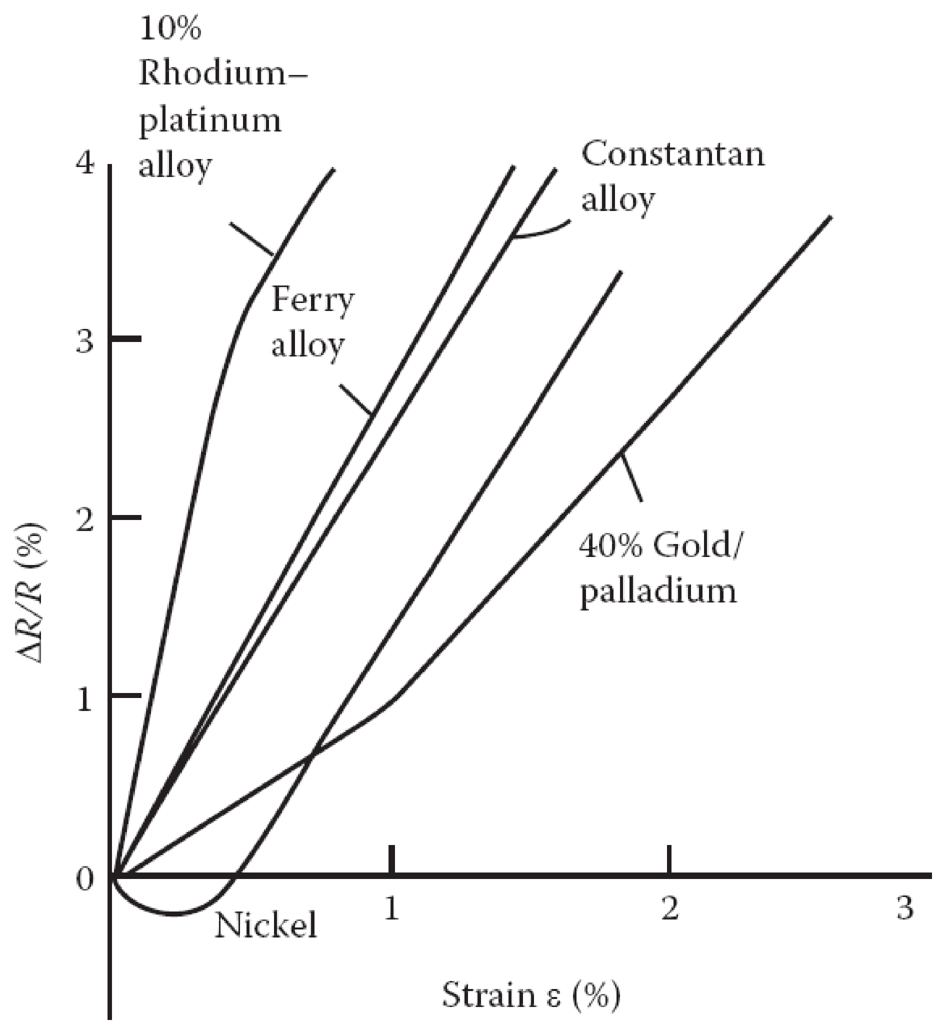
Temperature coefficient,

Resistivity,

Thermal stability.

Linearity of sensitivity.

اثر پیزورزیست



GF plots for various strain gauge element materials

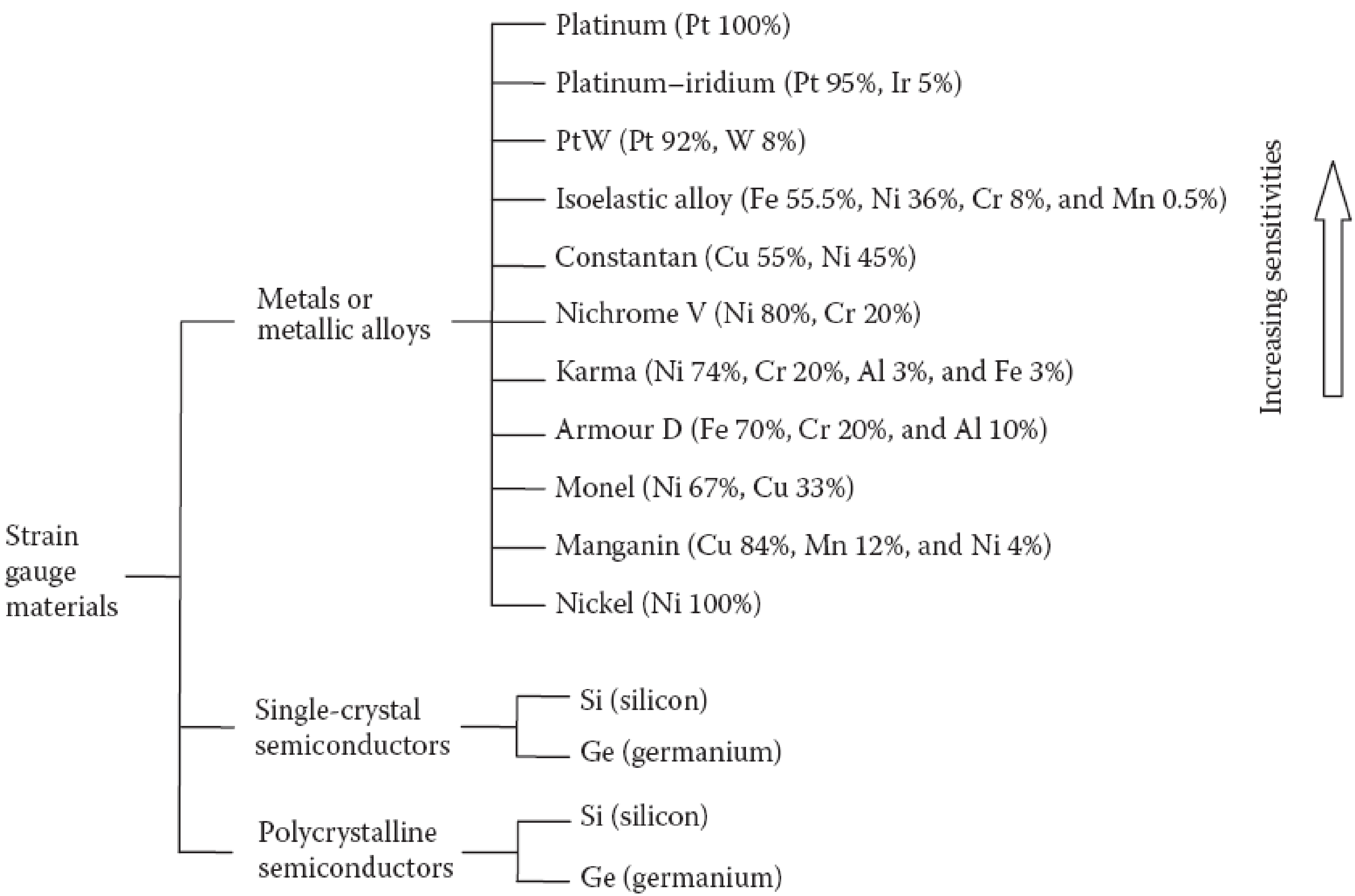
Gauge Factor and Ultimate Elongation for Several Materials

Material	Gauge Factor (GF) For Low Strain	Gauge Factor (GF) For High Strain
Copper	2.6	2.2
Constantan	2.1	1.9
Platinum	6.1	2.4
Silver	2.9	2.4
40% Gold/palladium	0.9	1.9

Typical GF Range of Main Types of Strain Gauges

Type of Strain Gauge	Gauge Factor (GF)
Metal foil	1 ~ 5
Thin-film metal	≈ 2
Bar semiconductor	80 ~ 150
Diffused semiconductor	80 ~ 200

Ebook: W. Y. Du "*Resistive, capacitive, inductive, and magnetic sensor technologies*", CRC Press, 2014.



Ebook: W. Y. Du "*Resistive, capacitive, inductive, and magnetic sensor technologies*", CRC Press, 2014.

Piezoresistive strain gauges are applied in measuring acceleration, force, torque, pressure, and vibration.

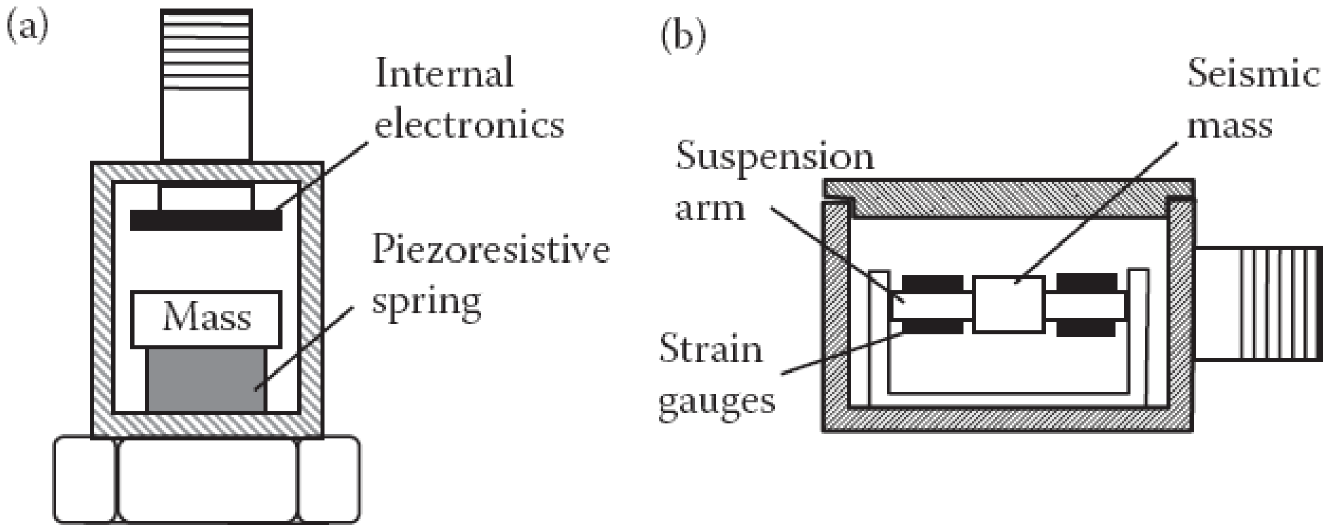
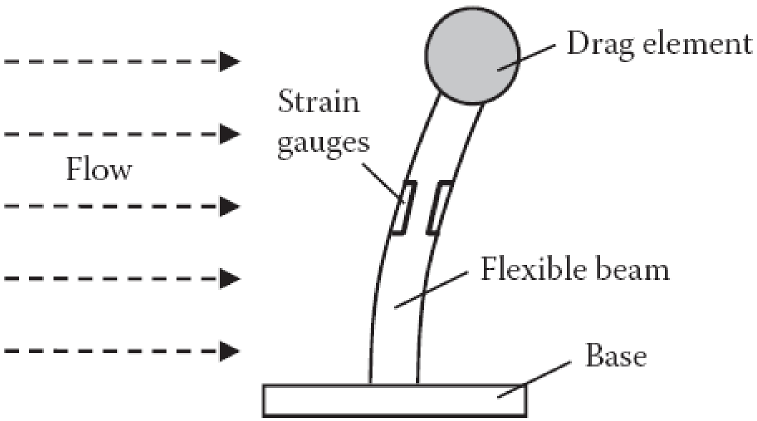


FIGURE 2.36 (a) Piezoresistive spring type accelerometer; (b) suspension arm type

کاربرد حسگر پیزو رزیستیو

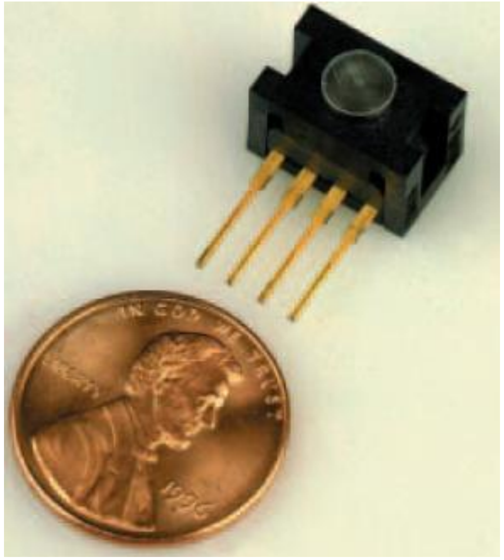


A piezoresistive flow rate sensor.

TABLE 2.18

Specifications of LPM 560 Micro Force Sensor

Load range: 0–1500 grams	Linearity: 22.5–25 grams
Hysteresis: 45–180 grams	Repeatability: 30–120 grams
Temperature range: −40 to 185°F	Output: 290–430 mV/FS
Bridge resistance: 5 kΩ	Excitation voltage: 10 VDC



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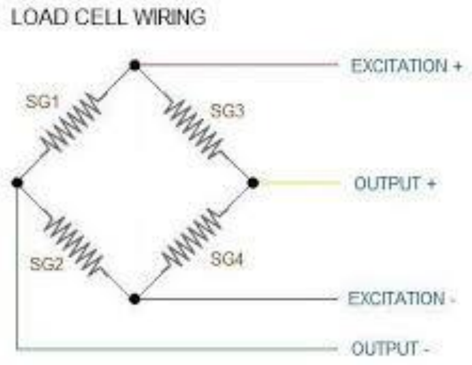
انواع نیروسنج Load cell







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